

Research Paper

Dual-scale flower detection and homography-guided cluster matching for robotic pollination in trellised kiwifruit orchards

Leilei He^a, Bo Jia^a, Juncai Huang^a, Rui Li^a, Spyros Fountas^c, Longsheng Fu^{a,b,*} ^a College of Mechanical and Electronic Engineering, Northwest A&F University, Yangling, Shaanxi, 712100, China^b School of Electronic Information and Artificial Intelligence, West Anhui University, Lu'an, Anhui, 237012, China^c Agricultural University of Athens, Athens, 11855, Greece

ARTICLE INFO

Keywords:

Flower detection
Dual-scale perception
Homography transformation
Flower cluster matching
Robotic pollination

ABSTRACT

The asynchronous flowering of kiwifruit presents a significant challenge for robotic pollination, which requires accurate flower detection and fine-grained phenological classification under complex orchard conditions. Existing perception systems are typically constrained by a single field of view, forcing a compromise between broad canopy coverage and detailed flower-level observation. To address this limitation, this study proposed a dual-scale perception framework that integrates far-view scanning, near-view refinement, and homography-guided cross-view flower cluster matching. A differentiated annotation strategy of flowers was designed for the two views, in which far-view detection enabled three classes based on flower openness. While near-view detection refined recognition into eight classes incorporating phenological phase and pollination status. Experiments were conducted to evaluate the impact of input size to YOLO26l on detection performance. The optimal configuration with an input size of 1280 achieved a mean average precision (mAP) of 0.902 and 0.935 for far-view and near-view flower detection, respectively. A cross-view flower cluster matching method was further developed by combining positional and appearance descriptors derived from homography transformation and colour space. The applied global-domain detection for matching achieved robust cluster matching with an accuracy of 94.83%, improving by 7.89% compared with local-domain detection for matching strategy. By integrating wide-area perception with precise flower-level classification, the proposed framework establishes a multi-scale perception pipeline that supports canopy-scale flower distribution analysis and cross-view target association. These capabilities provide a practical perception foundation for subsequent robotic operation planning and selective pollination in trellised kiwifruit orchards.

Nomenclature

Symbols	
α	Scale factor of histogram similarity score
β	Scale factor of mean colour similarity score
γ	Scale factor of position similarity score
Acc	Accuracy
AP	Average Precision
C	Centre point of bounding box to detected flower cluster
c'	Transformed coordinate of C
$d(i, j)$	Euclidean distance between C and c
d_{max}	Threshold for distance normalisation
e	Reprojection error threshold
F_1	Histogram similarity score
F_2	Mean colour similarity score
F_3	Position similarity score

(continued on next column)

(continued)

FC_d	The number of flower clusters that are detected in near-view images
FC_t	The number of correctly matched flower clusters in near-view images
H	Homography matrix
h	Histogram vector
i	The index of flower cluster in the near-view
j	The index of flower cluster in the far-view
mAP	Mean Average Precision
P	Precision
P_k	The homogeneous coordinates of a keypoint in the near-view image
P'_k	The homogeneous coordinates of a keypoint in the far-view image
R	Recall
s	Scale factor in homography
S	Similarity score of flower cluster between the near-view and far-view
TP	Truth Positives
FP	False Positives

(continued on next page)

* Corresponding author. College of Mechanical and Electronic Engineering, Northwest A&F University, Yangling, Shaanxi 712100, China.

E-mail address: fulsh@nwfau.edu.cn (L. Fu).<https://doi.org/10.1016/j.biosystemseng.2026.104561>

Received 25 March 2026; Received in revised form 29 July 2026; Accepted 30 July 2026

Available online 10 August 2026

1537-5110/© 2026 IAgRE. Published by Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

(continued)

Abbreviations	
BD	Bud
B _{FC}	Bounding box of detected flower cluster
BFMatcher	Brute-Force Matcher
BJ	Branch junction
BRISK	Binary Robust Invariant Scalable keypoints
CNN	Convolutional neural networks
DLT	Direct Linear Transformation
D _{FV}	Far-view dataset
D _{NV}	Near-view dataset
D _{FV-OPS}	Far-view dataset with overlap-partitioning strategy
FC	Flower cluster
FoV	Field of view
GDM	Global-domain detection for matching
imgsz	Size of the input image
HOF	Half open flower
LDM	Local-domain detection for matching
OCF	Occluded flower
OF	Open flower
OPF	Open pollinated flower
OPS	Overlap-partitioning strategy
OUF	Open unpollinated flower
RANSAC	Random Sample Consensus
SV	Stereo vision
ToF	Time-of-flight
UOF	Unopen flower
WF	Wilted flower

1. Introduction

Pollination is essential for ensuring fruit set and development in kiwifruit production. Although natural agents such as insects and wind have long contributed to crop pollination, their reliability is increasingly limited by declining pollinator populations and unstable climatic conditions (Garibaldi et al., 2014; Millard et al., 2023; Rodger et al., 2021). To compensate for these limitations, artificial pollination has been widely adopted in commercial orchards and has substantially improved fruit yield and marketability (Mazinani et al., 2023). However, this process depends heavily on skilled seasonal labour, which is becoming increasingly unsustainable at large scales under the growing shortage of agricultural labour (Abbate et al., 2021). The growing demand for efficiency, precision, and scalability therefore highlights the need for robotic pollination systems. Such systems are designed to replace repetitive manual operations with automated, perception-driven interventions, enabling more consistent pollination performance across heterogeneous crop structures and large-scale production systems (Xie et al., 2025; Zhang et al., 2025, 2026).

Robotic pollination systems can be categorised according to their operational scale, perception dependency, and carrier platform. Broad-area systems, commonly deployed on unmanned aerial vehicles or large mobile platforms, mainly rely on airflow-assisted or spray-based pollen dispersion to achieve rapid canopy coverage without explicit flower-level localisation (Zhang et al., 2026). Their effectiveness depends strongly on crop reproductive characteristics and canopy architecture. Crops such as walnut (Mazinani et al., 2023), pear (Xu et al., 2023), and date palm (Alyafei et al., 2022) generally exhibit relatively open canopies and partially wind-assisted pollination characteristics, whereas crops such as tomato require localised flower-level interaction (Hiraguri et al., 2023). Kiwifruit presents a more complex scenario with densely clustered flowers beneath trellis-supported canopies and highly asynchronous flowering stages. These biological and structural differences increasingly drive the need for precision-oriented robotic pollination systems with flower-level perception and targeted pollen delivery. Moreover, even broad-area pollination systems increasingly rely on visual perception for operation monitoring, flowering density estimation, and adaptive pollen regulation (Pandey et al., 2024). Consequently, robust visual perception and spatial flower understanding

are becoming fundamental prerequisites for vision-guided and target-oriented robotic pollination systems.

Advances in agricultural robotics have increasingly shifted from isolated perception algorithms toward integrated system-level solutions that jointly consider crop morphology, canopy architecture, robotic perception, and field operational efficiency. With the rapid advancement of deep learning, convolutional neural networks (CNNs) have become the predominant approach for object detection, offering superior robustness and adaptability across varying field conditions (Uddin et al., 2026; Williams et al., 2020; Xiao et al., 2025). Recent studies have reported promising results, with mean Average Precision (mAP) for kiwifruit flower detection exceeding 90%, and expanding beyond single-class detection to include flowering stage classification and operational status assessment, thereby supporting a broader range of agronomic management tasks (Li, Suo, et al., 2022, Li, Fu, et al., 2022; Liu & Li, 2025; Pan et al., 2024). To further enhance field-scale robotic operation, emerging approaches have also integrated geometric information from 3D point clouds and dynamic trajectory prediction for more precise target localisation and planning under complex unstructured conditions (Wang et al., 2025a, 2026). Despite these advances, most studies focus solely on local spatial object detection and task planning. This prevents robots from perceiving flower distribution at canopy scale and constrains their capability to support operation planning in canopy level (Gao et al., 2023). In addition, a restricted fields of view (FoV) constrain the plannable workspace, requiring frequent robot repositioning and increasing the risk of targets loss, thereby reducing applicability in extensive field environments (Gao et al., 2025).

Expanding the perceptual FoV is essential for enlarging the executable workspace and enabling coordinated canopy-level pollination planning. A straightforward solution is to increase the camera-to-canopy distance to cover a wider area in a single image. However, this reduces the pixel density of individual flowers, compromises detection reliability, and further degrades depth-estimation accuracy because methods such as time-of-flight (ToF) and stereo vision (SV) have limited resolution and greater uncertainty at long ranges (Wang et al., 2022). Another common strategy to achieve a larger view involves sequentially capturing and stitching overlapping images from a moving camera, which has been widely used in detection-based object counting to combine global context with local details across multiple perspectives (He et al., 2022; Mirhaji et al., 2021; Zhou et al., 2022). Although this segmented perception approach enriches environmental information, it is prone to stitching errors by the inherent complexity of illumination variation, occlusion, and canopy deformation in field orchards (Du et al., 2024; Zhu et al., 2025). Furthermore, frequent image acquisition requires repeated robotic arm movements, introducing redundant motion that increases task duration and reduce overall pollination efficiency. Therefore, developing an integrated perception strategy that balances wide-area coverage with accurate local recognition is crucial for achieving efficient and reliable robotic pollination.

Multi-scale visual perception provides a promising strategy for improving spatial coverage and detection robustness by combining wide-angle and close-up observations in field orchards. Active perception has been proposed as a sequential strategy that moves progressively from a wide overview to close-up inspection for refined perception (Magalhães et al., 2022; Yang, Li, et al., 2024). Although effective in enlarging the observable scene, this sequential refinement struggles to balance FoV, detection precision, and localisation accuracy when relying on a single camera (Wang, Burusa, et al., 2025). SV, the practical solution for field localisation, particularly requires a large baseline to improve depth resolution at long distances, while a small baseline is necessary at short ranges to prevent excessive disparity and matching points loss (Diaz-Ramirez et al., 2025). To overcome these limitations, multi-camera systems that integrate far-view and near-view sensors have emerged as an effective solution, providing complementary perspectives across different spatial scales for objects detection in agricultural tasks such as harvesting (Rybak et al., 2025; Villacrés &

Vougioukas, 2024; Yang, Noguchi, & Hoshino, 2024). Nevertheless, existing studies have paid limited attention to cross-view identity association for the same target observed from different viewpoints. This issue is particularly important for robotic pollination, because the same flower may otherwise be repeatedly pollinated.

In this study, a far-near view cooperative perception framework was proposed for robotic pollination in trellised kiwifruit orchard. Wide-field far-view images were used to identify flower clusters across the canopy, whereas near-view images enabled fine-grained classification of individual flowers including phenological stage and pollination status. To connect the two perception scales, a cross-sensor and cross-view flower cluster matching method was developed by integrating appearance features and position similarity to establish and maintain consistent instance identities. The proposed framework establishes consistent flower cluster identities across observation scales and supports canopy-level operation planning, selective flower-level pollination, and reduction of redundant pollination.

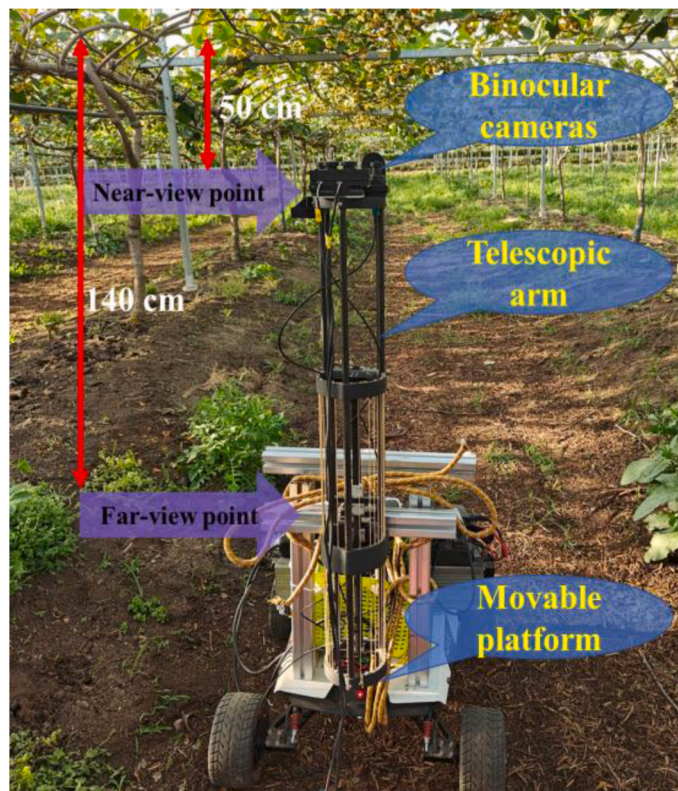
2. Materials and methods

2.1. Image acquisition

Images of kiwifruit flowers were collected in a commercial of the International Kiwifruit Innovation Orchard, Yangling, Shaanxi Province, China (34°17' N, 108°2' E; approximately 504 m altitude). This orchard adopts a standardised pergola trellis cultivation system that provides a relatively uniform canopy structure and enables imaging from beneath the canopy to support flower detection and precision pollination. To ensure morphological representativeness, three widely cultivated local varieties of the *Actinidia deliciosa*, 'Cuixiang', 'Xuxiang', and 'Nongdayuxiang', were selected for data collection. Image acquisition was conducted across the entire flowering season, covering bud stage, early

flowering, anthesis, and senescence stage. The number of 552 far-view images and 565 near-view images were captured under natural light conditions, encompassing variations in illumination intensity to enhance environmental diversity.

To capture both canopy-level coverage and detailed flower morphology, a dual-scale image acquisition strategy was employed, consisting of far-view and near-view imaging. For far-view acquisition, a binocular camera (ALG Stereo Camera L) with a resolution of 3840×2160 pixels and an FoV of 120° was positioned at approximately 1.4 m below the canopy, enabling wide-field observation of flowers across the trellis. Near-view acquisition was performed at a closer distance of approximately 0.5 m from the canopy to enhance the visibility of individual flowers, using a binocular camera (JIERUIWEITONG SH500) with a resolution of 2560×1440 pixels and an FoV of 100° . All cameras were mounted on a movable platform (SCOUT MINI, Agilex Robotics, China) with a telescopic arm to ensure a relatively stable distance from canopy during image acquisition. The platform and example images captured from different perspectives are shown in Fig. 1. Before image acquisition, all binocular cameras were calibrated using the Stereo Camera Calibrator toolbox from MATLAB R2025a (MathWorks, 2025) with a checkerboard-based calibration procedure. The calibration results were used primarily for image distortion correction and epipolar rectification of binocular image pairs. In this study, only corrected left-view images were utilised for subsequent flower detection and cross-view matching, while homography estimation was performed from image feature correspondences without directly incorporating calibrated camera parameters into the transformation process. All images are available at <https://github.com/Smart-Ag/Far-Near-kiwiflower-detection-matching-images>.



(a)



(b)



(c)

Fig. 1. (a) Mobile platform for image acquisition equipped with a telescopic arm, and example images that collected from (b) the far-view and (c) near-view.

2.2. Dataset preparation

2.2.1. Data annotation

According to established property of image acquisition, data annotation followed the principle of tailoring annotation granularity to different scales. For far-view images, which primarily served to expand the FoV for pollination robots, without performing the excessive details of individual flowers. Therefore, a multi-class annotations emphasised by major reproductive of flowering stages including unopen flower (UOF), open flower (OF), and flower cluster (FC). For near-view images, a finer annotation scheme was used to capture pollination status and occlusion state, supporting the detailed detection required for selective pollination and robot accessibility. Specifically, the multi-class objects annotation of near-view images include bud (BD), half open flower (HOF), open unpollinated flower (OUF), open pollinated flower (OPF), wilted flower (WF), occluded flower (OCF), flower cluster (FC), and branch junction (BJ) that assisted for flower cluster definition as a structural reference (Li, Fu, et al., 2022). All annotations were performed manually by open-source Labellmg (<https://github.com/tzutalin/labellmg>) in PASCAL VOC format. Representative annotation examples were shown in Fig. 2, where flowers of different classes are annotated with rectangular boxes.

2.2.2. Data augmentation

To enhance dataset diversity and improve model generalisation under orchard conditions, data augmentation was applied uniformly to the far-view and near-view images. The two views were used to construct separate datasets for training and evaluating their corresponding detection models independently. Within each dataset, all images were first divided into training and validation sets at a ratio of 8:2, after which augmentation was applied only to the training set to maintain independence between model training and evaluation. Data augmentation strategies, including fixed-angle rotations of 90°, 180°, and 270°, horizontal flipping, and vertical flipping, were applied to the divided training sets. Motion blur was also introduced to mimic natural disturbances such as wind-induced branch movement. All transformations were applied simultaneously to annotation files and images, preserving their consistency. After augmentation, original training set

was expanded sixfold, and augmented datasets were referred to as the far-view dataset (D_{FV}) and near-view dataset (D_{NV}), respectively.

2.2.3. Overlap-partitioning strategy for far-view detection

For D_{FV} , the large FoV captures a wide canopy area, but individual flowers occupy only a limited number of pixels. This small-object scale poses a challenge for deep learning models, as resizing high-resolution images to a fixed network input size (typically set to 640×640 pixels) often causes the loss of fine details and degrades detection accuracy. Increasing the input size can mitigate this issue by preserving more pixel-level information. However, continuous enlargement of the input size does not yield proportional improvements in model performance or training loss reduction, since the feature extraction capacity of networks and computational constraints impose inherent limits on performance gains. To address this issue, an overlap-partitioning strategy (OPS), which partitions high-resolution images into multiple patches with a controlled overlap rate was utilised. This process preserves small-object resolution while maintaining compatibility with a required input size of network (Dang et al., 2025). Specific process of OPS applied in this study has been described detailly in the previous study of He, Wei, et al. (2025). In brief, each far-view image was partitioned into multiple overlapping sub-images using a sliding-window strategy, where the overlap region between adjacent patches was determined according to the predefined overlap ratio relative to the patch size. According to the scale distribution of kiwifruit flowers, the overlap rate and resolution of partitioned patches were set to 35% and 1280×1280 pixels, respectively. Each original far-view image generated 18 overlapping patches, resulting in 7938 images of training set and 1980 images of validation set that integrated as far-view dataset with OPS (D_{FV-OPS}) to ensure a balance between accuracy and computational efficiency.

2.3. Objects detection based on deep learning

2.3.1. Network selection

The state-of-the-art You Only Look Once version 26 (YOLO26) (<https://github.com/ultralytics/ultralytics>) was adopted in this study for kiwifruit flower detection, offering significant improvements in

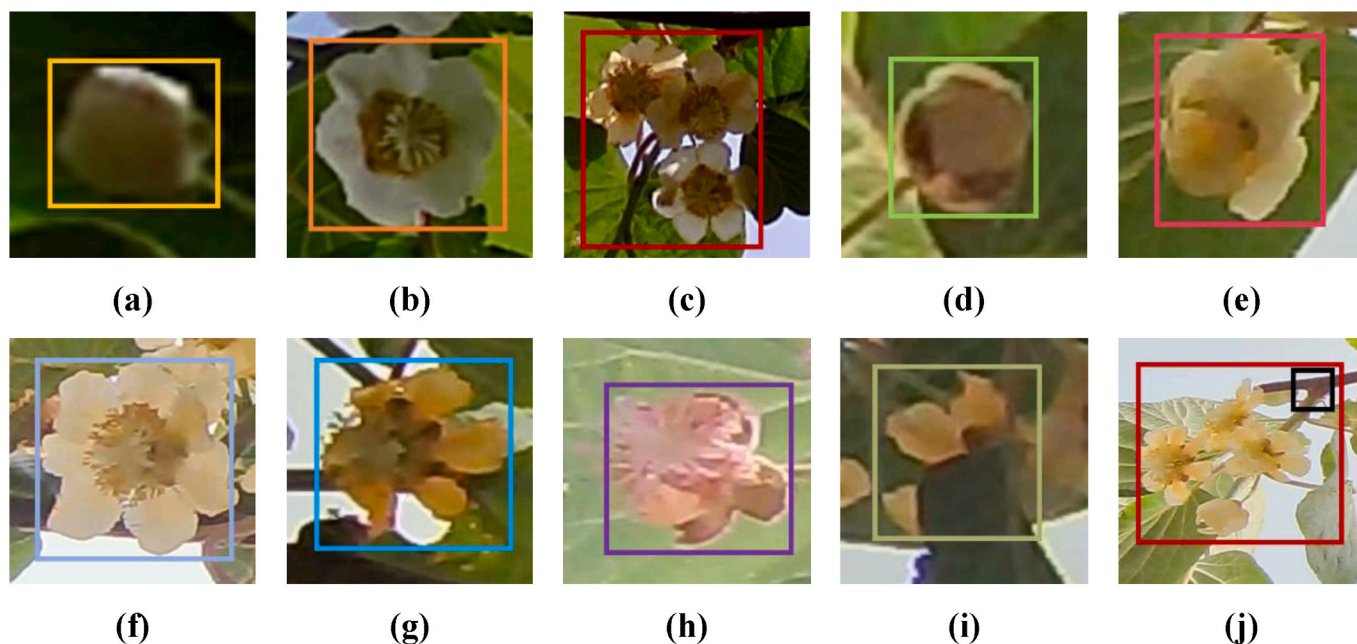


Fig. 2. Examples of multi-class annotation for kiwifruit flowers, including (a) UOF, (b) OF, and (c) FC in far-view images with (d) BD, (e) HOF, (f) OUOF, (g) OPF, (h) WF, (i) OCF, and (j) FC (red box) assisted by BJ (black box) in near-view images. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

computational efficiency, detection precision, and hardware adaptability. The core architecture of YOLO26 is illustrated in Fig. 3. Moving away from traditional architectures, YOLO26 introduces an end-to-end Non-Maximum Suppression-free (NMS-free) design and removes the Distribution Focal Loss (DFL) module, which minimises inference latency jitter and maximises deployment compatibility on edge devices (Hidayatullah et al., 2026; Sapkota et al., 2026). Furthermore, it integrates Progressive Loss (ProgLoss) and Self-Training Anchor Loss (STAL) mechanisms that dynamically adjust network attention, making it highly effective for detecting small targets in heavily occluded scenarios. The incorporation of the novel MuSGD optimiser alongside pre-trained weights accelerates convergence and enhances generalisation, which is particularly beneficial when working with relatively limited datasets such as flower images. In this study, the YOLO26 large (YOLO26l) variant was employed for kiwifruit flower detection, as it offers a favourable balance between detection accuracy and computational demand. Compared with smaller versions, YOLO26l provides enhanced feature representation capacity, which is particularly

important for capturing small, densely clustered, and partially occluded flower targets within complex orchard backgrounds. This improved representational ability, together with its optimised inference pipeline, enables stable real-time performance under edge-computing constraints, making it suitable for downstream robotic pollination tasks.

2.3.2. Network training

Flower detection models were trained using YOLO26l on prepared datasets of D_{FV} , D_{FV-OPS} , and D_{NV} . The training process was carried out on a workstation equipped with an Intel Xeon Gold 5218 CPU, 128 GB of RAM, and an NVIDIA RTX 4090 GPU with 24 GB memory, running a 64-bit Windows 10 operating system. The software environment consisted of Python 3.10.14, PyTorch 2.2.2, CUDA 12.1, and cuDNN 8.8.1. To improve model generalisation, transfer learning was employed by initialising the network with pre-trained weights from the Common Objects in Context dataset (<https://docs.ultralytics.com/zh/models/YOLO26>). For each dataset, three input sizes of 640, 960, and 1280 were systematically evaluated to investigate the influence of spatial

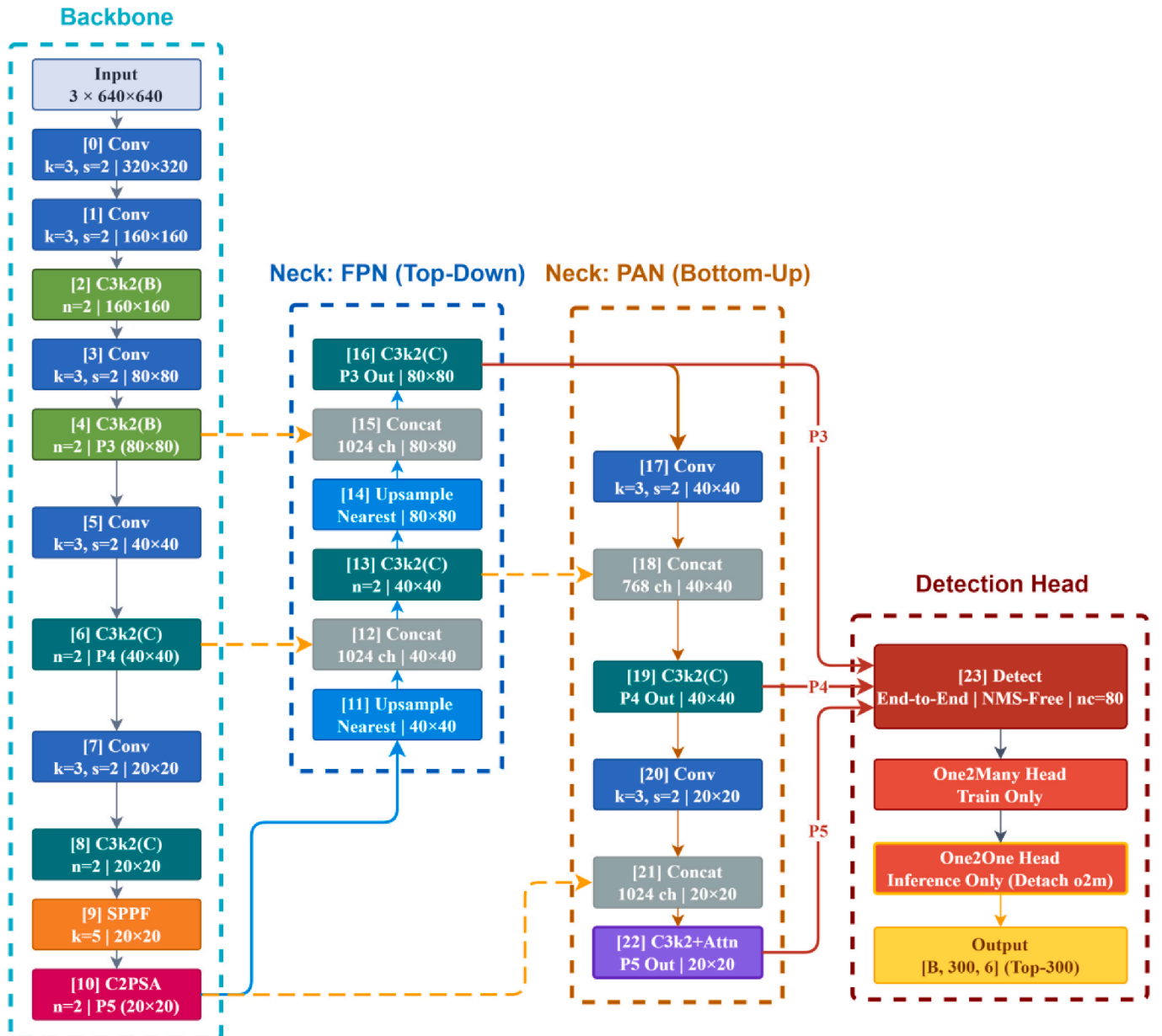


Fig. 3. Core architecture diagram of YOLO26.

resolution on detection accuracy and computational efficiency. All models were trained under identical hyperparameter settings to ensure fair comparison across datasets. A batch size of 4 was used due to GPU memory limitations, with an initial learning rate of 0.001, a final learning rate factor of 0.01, and a weight decay of 0.0005. Each model was trained for up to 500 epochs, incorporating early stopping with a patience of 50 epochs to mitigate overfitting.

2.4. Flower cluster matching

After flower detection, a hybrid-feature fusion matching framework was developed to establish robust correspondences between flower clusters observed from different views. The framework integrates homography-based geometric alignment with appearance-based feature representation to jointly characterise spatial consistency and photometric similarity across views. Appearance descriptors were constructed from colour statistics of detected flower clusters, while geometric descriptors were derived from homography-transformed spatial relationships. Final correspondences were obtained through bipartite assignment optimisation. The following pseudocode summarises the overall matching workflow between far-view and near-view images.

Steps	Algorithm: Flower cluster matching
	Input: Far-view image I_f with detected flower cluster boxes $B_{FC,f}$ Near-view image I_n with detected flower cluster boxes $B_{FC,n}$ Parameters: detector type, matcher type, threshold τ ($\tau = 0.4$)
	Output: Set of matched cluster pairs with confidence scores
S1:	Load I_f and I_n ; convert $B_{FC,f}$ and $B_{FC,n}$ from YOLO format to pixel coordinates
S2:	Initialise BRISK detector and BFMatcher
S3:	Detect keypoints in I_f and I_n
S4:	Match keypoints using BFMatcher; filter by a ratio test
S5:	Estimate homography matrix H with RANSAC
S6:	For each bounding box in $B_{FC,f}$ and $B_{FC,n}$: Extract image patch Compute HSV colour histogram and mean RGB colour Record cluster centre (apply H to far-view centres)
S7:	Construct similarity matrices: histogram, mean colour, and spatial position (if H exists)
S8:	Combine similarities into unified score matrix S
S9:	Apply Hungarian algorithm on S to enforce one-to-one assignment
S10:	For each assignment, if $S(i, j) \geq \tau$, add $(i, j, S(i, j))$ to match set
S11:	Return final matched cluster pairs

2.4.1. Homography matrix estimation

As a trellised crop whose branches are constrained by supporting wires, kiwifruit flower clusters lie approximately on a locally planar canopy surface. Therefore, the geometric discrepancies between cross-view observations are primarily caused by variations in camera pose, distance, and orientation. In this case, homography matrix was estimated to capture the projective relationship between far-view and near-view images. Homography estimation between cross-view images followed a feature-based geometric alignment pipeline. First, keypoints and descriptors were extracted using the Binary Robust invariant scalable keypoints (BRISK) (Leutenegger et al., 2011), parameterised with a detection threshold of 50 and 3 octaves to optimise for scalability and invariance. Greyscale conversions of cross-view images were utilised for feature detection. Descriptor matching was performed using the Brute-Force Matcher (BFMatcher) with Hamming distance and k -nearest neighbour matching ($k = 2$). Lowe's ratio test with a threshold of 0.8 was subsequently applied to reject ambiguous correspondences, and the number of retained matches was capped at 300 to improve computational stability and efficiency. Subsequently, the Random Sample Consensus (RANSAC) algorithm was applied to the matched keypoints for homography estimation by Direct Linear Transformation (DLT), using a reprojection error threshold of 3.0 pixels and a confidence level of 0.995 to further remove geometric outliers. To reduce computational

load, cross-view images were initially downsampled by a ratio of 0.5 for feature extraction and matrix estimation, after which the estimated matrix was restored mathematically to the original resolution. The homography transformation between cross-view images is defined in Eq. (1) and Eq. (2), and the reprojection error of a candidate homography is defined in Eq. (3).

$$sP'_k = HP_k \quad (1)$$

$$H = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \quad (2)$$

$$e = \|P'_k - \hat{P}'_k\|_2 \quad (3)$$

where, $P_k = (x_k, y_k, 1)^T$ refers to the homogeneous coordinates of a keypoint in the near-view image; $P'_k = (x'_k, y'_k, 1)^T$ refers to the homogeneous coordinates of a keypoint in the far-view image; k refers to the index of keypoint pairs; H refers to the non-singular homography matrix estimated from keypoint pairs correspondences, s refers to the scale factor; $\hat{P}'_k$ refers to the projection of P_k under the estimated homography; and matches with e below the threshold (set to 3.0 pixels) were classified as inliers.

Two strategies for keypoints detection based on image domain selection were considered of global-domain detection for matching (GDM) and local-domain detection for matching (LDM). GDM extracts feature from the entire image pair to provide a rich set of keypoints, whereas LDM extract features only from the flower clusters detected by trained flower detection model to minimise background interference.

2.4.2. Feature representation and matching

A hybrid feature representation approach was applied to combine position and appearance descriptors for discriminative and resilient flower cluster matching under diverse environmental conditions. Specifically, position and appearance descriptors were constructed to each bounding box of detected flower cluster (B_{FC}). Appearance descriptors include a histogram from the HSV (Hue, Saturation, Value) colour space for distribution analysis and a mean vector from RGB colour space for summary statistics, both derived from the regions of B_{FC} after a 5 pixels expansion and boundary clipping to incorporate limited surrounding contextual information while reducing excessive background interference. Expanded B_{FC} regions were converted into HSV colour space, where the Hue and Saturation channels were utilised independently from the Value channel to improve robustness against illumination variations. The colour histogram was then computed using 8 bins for the Hue channel and 6 bins for the Saturation channel following standard HSV conversion. The resulting 2D histogram is normalised using L2 normalisation before flattening into a 48-dimensional feature vector, which reduces the influence of varying bounding-box sizes between cross-view observations. Complementarily, the mean colour is extracted as the average RGB value of the clipped B_{FC} region, normalised to the range of [0, 1] using Min-Max scaling for consistency in similarity computation. Appearance descriptors of flower clusters from cross-view images were subsequently utilised for cosine-based similarity calculations during matching, as defined in Eq. (4) and Eq. (5).

$$F_1(i, j) = 1 - \cos(h_i, h_j) \quad (4)$$

$$F_2(i, j) = 1 - \cos(m_i, m_j) \quad (5)$$

where, F_1 refers to the histogram similarity score of flower cluster between the far-view and near-view; i refers to the index of flower cluster in the near-view; j refers to the index of flower cluster in the far-view; h_i refers to the histogram vector of the i th flower cluster in the near-view; F_2 refers to the mean colour similarity score of flower cluster between the far-view and near-view; m_i refers to mean colour vectors of the i th flower cluster in the near-view.

The position descriptor involves describing the spatial relationships of flower cluster between different views. First, the centre of B_{FC} from the near-view was transformed to far-view coordinate system using the estimated homography matrix H . After that, the positional similarity was derived from the Euclidean distance between B_{FC} centres after homography transformation. The transformation of box centres in cross-views is defined in Eq. (6), while positional similarity is defined in Eq. (7) and Eq. (8).

$$c' = \frac{H \cdot [C, 1]^T}{[H \cdot [C, 1]^T]_z} \quad (6)$$

$$d(i, j) = \sqrt{(x_j - x_i)^2 + (y_j - y_i)^2} \quad (7)$$

$$F_3(i, j) = \max(0, 1 - \min(1, d(i, j) / d_{\max})) \quad (8)$$

where, C refers to the centre point of B_{FC} ; $[\cdot]_z$ refers to the z-component for perspective normalisation; c' refers to the transformed coordinate of C ; (x_j, y_j) refer to the coordinates of C_j from the far-view; (x_i, y_i) refer to the coordinates of c'_i , which is the transformed coordinate of C_i from the near-view; $d(i, j)$ refers to the Euclidean distance between C_j and c'_i ; F_3 refers to the position similarity score of flower cluster between the far-view and near-view; $d_{\max}$ refers to the threshold used for distance normalisation, which was set to 0.1 times the pixel width of far-view image in the x-axis.

After representative features of flower clusters were extracted from the far-view and near-view images, cross-view correspondences between clusters were established. For each pair (c'_i, C_j) , a similarity score $S(i, j)$ was computed by integrating three feature-based metrics, as defined in Eq. (9). Direct pairwise association may suffer from ambiguity when multiple clusters exhibit similar appearance or spatial proximity. To optimise the matching process, the Hungarian algorithm was employed to ensure a one-to-one assignment that maximised total similarity score while preventing multiple allocations to the same cluster. This combinatorial optimisation formulates the bipartite matching problem, in which near-view flower clusters are assigned to far-view clusters, as the search for the permutation with the highest aggregate score.

$$S(i, j) = \alpha F_1(i, j) + \beta F_2(i, j) + \gamma F_3(i, j) \quad (9)$$

where, $S(i, j)$ refers to the similarity score; α , β , and γ refer to the scale factor of constructed feature descriptors, which were specifically set to 0.2, 0.2, and 0.6 optimised by grid search, respectively.

2.5. Evaluation metrics

In this study, performance evaluation consisted of multi-class flower detection and flower cluster matching. For flower detection, the far-view and near-view models were assessed using Precision (P_n), Recall (R_n), Average Precision (AP_n), mAP, and average detection speed, following standard object-detection evaluation protocols (Everingham et al., 2010; Lin et al., 2014). P_n and R_n describe detection correctness and completeness, respectively, while AP_n and mAP summarise class-level and overall detection performance based on the precision-recall curve. The formulas for these evaluation metrics are provided in Eq. (10) to Eq. (13).

$$P_n = \frac{TP_n}{TP_n + FP_n} \quad (10)$$

$$R_n = \frac{TP_n}{TP_n + FN_n} \quad (11)$$

$$AP_n = \int_0^1 P_n(R_n) dR_n \quad (12)$$

$$mAP = \frac{1}{m} \sum_{n=1}^m AP_n \quad (13)$$

where, n refers to the index of specific object class; TP_n (True Positives) refers to the number of detections from the class correctly identified by the model; FP_n (False Positives) refers to the number of detections mistakenly assigned to the class; FN_n (False Negatives) refers to object from the class that the model failed to detect; and m refers to the total number of object classes.

Flower cluster matching of cross-view was evaluated using accuracy (Acc), defined as the proportion of near-view flower clusters correctly matched to their corresponding far-view clusters, as shown in Eq. (14).

$$Acc = \frac{FC_r}{FC_d} \times 100\% \quad (14)$$

where, FC_d refers to the number of flower clusters that are detected in near-view images; FC_r refers to the number of correctly matched flower clusters in near-view images.

3. Results and discussion

3.1. YOLO26l for far-view flower detection

The performance of YOLO26l for far-view flower detection was systematically evaluated under different training strategies and input resolutions. The curves of train/cls_loss for far-view detection models exhibited a sharp decline before stabilising, with early stopping preventing completion of 500 epochs, as shown in Fig. 4. Models trained with input size (defined as the image size (*imgsz*) of hyperparameter for model training) of 1280 achieved the lowest baseline and fastest convergence among different datasets. YOLO26l trained on D_{FV-OPS} obtained lower losses than the corresponding D_{FV} models, reflecting reduced training complexity by applied OPS. For D_{FV} , increasing the input size significantly improved flower detection performance across P , R , and mAP. Specifically, a progressive mAP improvement from 0.847 at an image size of 640 to 0.902 at 1280, reflecting that a large input size preserved finer flower structures and mitigated small-object feature loss by providing richer spatial details during model training. By contrast, models trained on D_{FV-OPS} exhibited only marginal gains across different input sizes, with mAP values remaining stable between 0.874 and 0.892, as shown in Table 1. This suggests that once image partitioning alleviates resolution constraints by preserving localised pixel density, further increasing the network input size contributes little to additional accuracy improvement. Thus, the OPS strategy effectively compensated for scale-induced degradation caused by standard resizing, but its benefit may saturate when sufficient spatial information is already preserved or when excessive contextual fragmentation is introduced (Akyon et al., 2022).

From the perspective of computational performance, increasing the input size led to a progressive rise in inference time for both training strategies. For the model trained on D_{FV} , detection speed decreased from 31.1 ms at an *imgsz* of 640 to 32.7 ms at 1280, which is consistent with a greater computational demand of processing larger inputs. A similar near-linear trend was observed for the model trained on D_{FV-OPS} , where detection speed declined from 30.4 ms at an *imgsz* of 640 to 31.5 ms at 1280. However, these per-patch values do not represent the total processing cost of OPS, because each original image was divided into 18 overlapped patches that required separate inference and result merging. This limitation restricts its applicability to online analysis or scenarios in which efficiency is prioritised over marginal precision gains (Xu et al., 2025). Consequently, OPS is advantageous when flowers occupy too few pixels after full-image resizing and preservation of small-target information is prioritised. Full-image inference is more appropriate when sufficient flower details can be retained within the available computational budget. Under the present imaging conditions, DFV with an *imgsz*

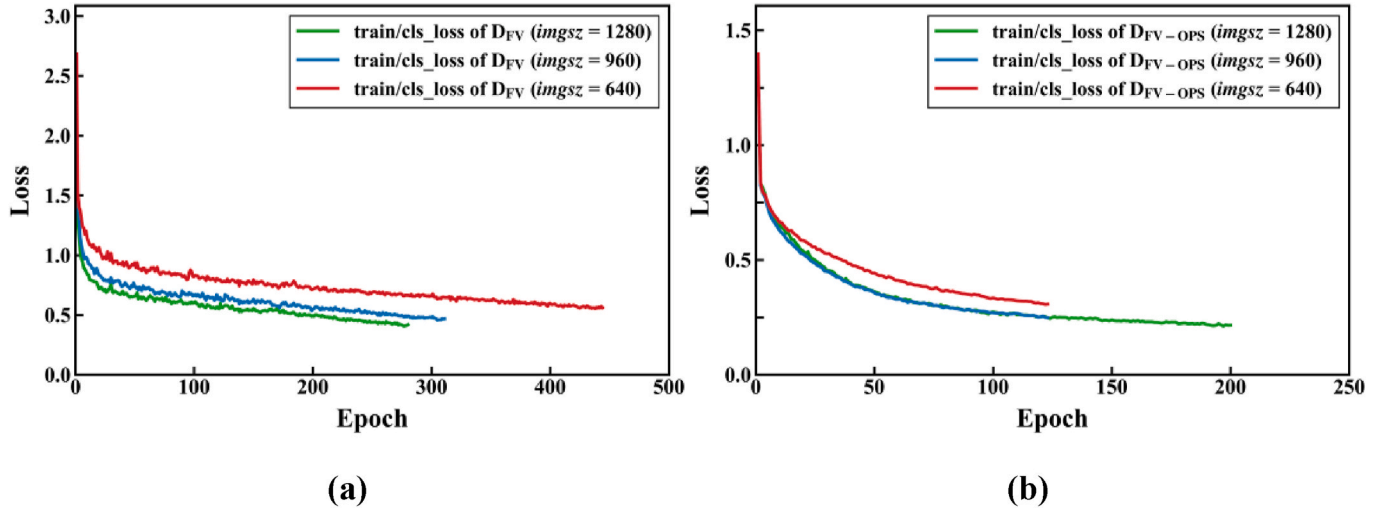


Fig. 4. Train/cls_loss curves for models trained on (a) D_{FV} and (b) D_{FV-OPS} .

Table 1

Performance of YOLO26l based on different datasets for far-view flower detection.

Datasets	Imgsz	P	R	mAP	Speed (ms/image)	Instances
D_{FV}	640	0.845	0.780	0.847	31.1	15,855
	960	0.857	0.835	0.893	32.0	
	1280	0.867	0.842	0.902	32.7	
D_{FV-OPS}	640	0.839	0.836	0.892	30.4	40,286
	960	0.831	0.829	0.887	30.3	
	1280	0.844	0.828	0.874	31.5	

of 1280 provided the most favourable balance between flower detectability and processing efficiency and was therefore selected for subsequent far-view detection and cross-view matching.

The YOLO26l model trained on D_{FV} with an *imgsz* of 1280 achieved robust detection of far-view flower classes. As shown in Table 2, the model achieved an overall *P* of 0.967, *R* of 0.842, and *mAP* of 0.902, indicating a strong balance between detection accuracy and stability. Among all individual classes, OF achieved the highest accuracy with an *AP* of 0.936 and *R* of 0.918, suggesting that its distinct shape and texture facilitated consistent detection even under canopy occlusion. Representative OF detections under occluded canopy conditions are illustrated in Fig. 5a. In contrast, UOF recorded slightly lower *P* of 0.822 and *AP* of 0.869, primarily due to its high morphological heterogeneity. This class includes both ellipsoidal flower buds and partially opened blossoms with irregular shapes, causing greater intra-class variation and several missed detections, as shown in Fig. 5b. In addition, the green buds tightly wrapped by sepals closely resemble canopy leaves in colour, thereby increasing misclassification and reducing detection accuracy, as shown in Fig. 5c. Similar patterns have been reported in tasks ranging from leaf species classification (Elbasi et al., 2024) to the detection of flowers and fruits in strawberry seedlings (Bai et al., 2024), where intra-class heterogeneity and background similarity significantly affect model precision. The FC class of the far-view achieved an intermediate performance with an *AP* of 0.877, reflecting moderate difficulty in

Table 2

Specific results of YOLO26l trained on D_{FV} with an *imgsz* of 1280.

Classes	P	R	AP	mAP
UOF	0.822	0.872	0.869	0.902
OF	0.936	0.918	0.960	
FC	0.843	0.827	0.877	
All	0.967	0.842	-	

distinguishing dense flower clusters from overlapping foliage. Overall, these results demonstrate that YOLO26l trained with the proposed configuration effectively captures category-specific features in the far-view while maintaining reliable generalisation under field conditions.

3.2. YOLO26l for near-view flower detection

Input size played an important role in the performance of YOLO26l for fine-grained multi-class flower detection in the near-view. As illustrated in Fig. 6, the train/cls_loss curves converged smoothly across all configurations, and an *imgsz* of 1280 achieved the lowest and most stable loss, indicating better feature learning and more stable optimisation during training. The performance metrics summarised in Table 3 show a steady improvement with increasing input size. When *imgsz* increased from 640 to 1280, *mAP* improved from 0.918 to 0.935 on D_{NV} . *P* and *R* showed similar gains, increasing from 0.911 to 0.891 to 0.927 and 0.905, respectively. These results confirm that larger input sizes enhance feature fidelity for small floral structures without obvious overfitting. The improvement by increased *imgsz* from 960 to 1280 was relatively limited, suggesting that most near-view flowers already retained sufficient spatial detail at an *imgsz* of 960. Nevertheless, detection time increased by only 0.8 ms over this interval. An *imgsz* of 1280 was therefore adopted when fine-grained flower classification was prioritised, whereas an *imgsz* of 960 may provide a more economical alternative for systems with stricter computational and GPU memory constraints.

The YOLO26l model trained on D_{NV} with an *imgsz* of 1280 achieved strong overall detection performance, demonstrating reliable generalisation across multiple flower classes. The model reached an overall *P* and *R* of 0.927 and 0.905, respectively, indicating a balanced capability for accurate recognition and low omission rates, as shown in Table 4. Among the individual classes, BD achieved the highest detection accuracy with an *AP* of 0.973, followed closely by OPF and FC with *AP* values of 0.972 and 0.967, respectively. These results demonstrate that higher-resolution inputs preserve fine-grained floral features, which are critical for distinguishing closely related flower classes. In contrast, HOF exhibited the lowest *AP* value of 0.886, which can be attributed to its intermediate growth morphology between buds and fully opened flowers, leading to higher intra-class variability (Li, Suo, et al., 2022). This aligns with the challenges faced in identifying UOF phenological stages from the far-view. BJ also showed a relatively lower *AP* of 0.899 with a *P* of 0.912 and *R* of 0.874, attributable to its frequent overlap with adjacent flower clusters. As shown in Fig. 7, due to unclear cluster

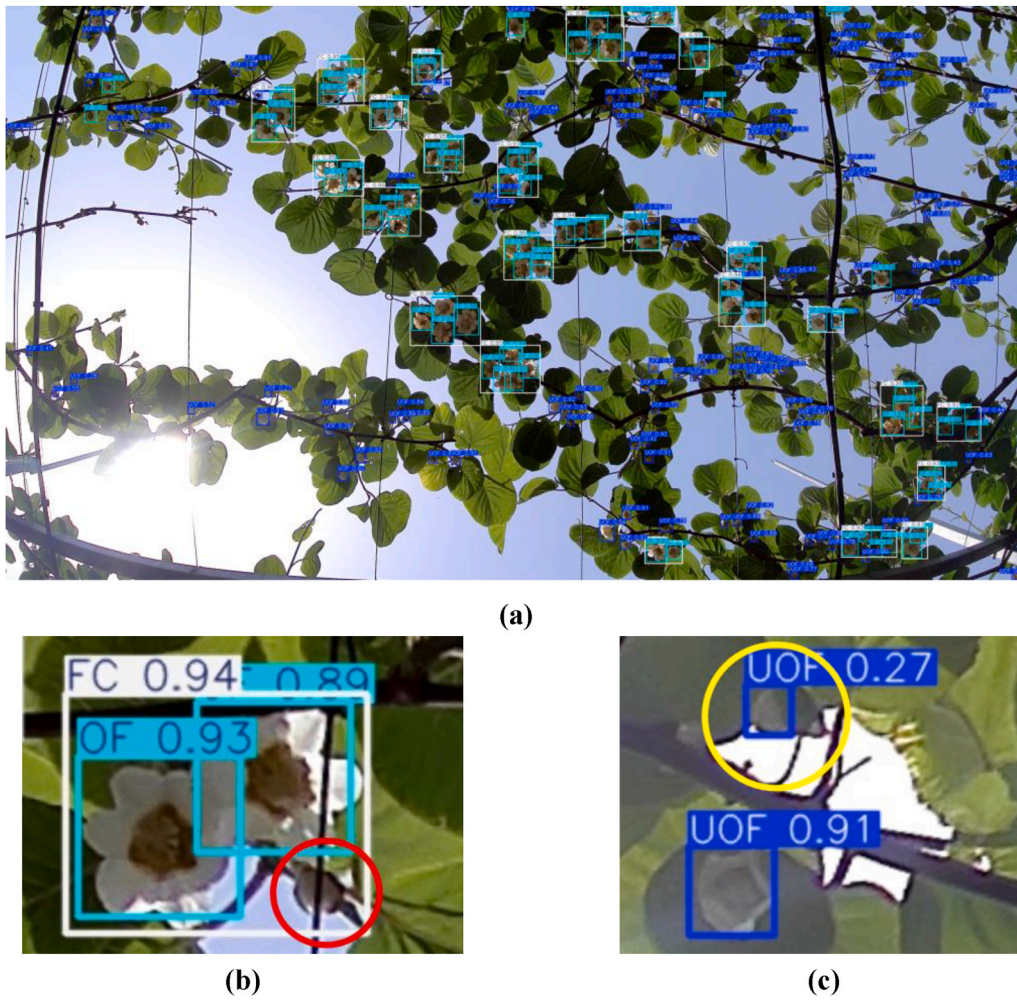


Fig. 5. Far-view detection examples from YOLO26l configured with an *imgsz* of 1280. (a) Canopy-level detection of kiwifruit flower; (b) Miss detection of UOF marked with a red circle; (c) Mistake detection of UOF marked with a yellow circle. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

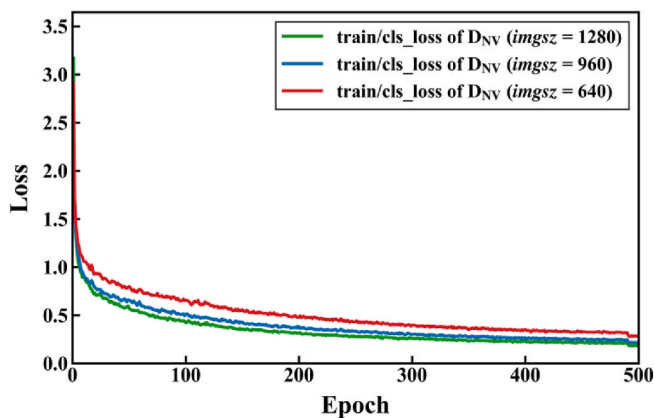


Fig. 6. Train/cls_loss curves for near-view models configured with different *imgsz*.

boundaries and occlusion within dense flower areas, several overlapping bounding boxes occasionally appear. Therefore, a post-processing step based on intersection-union ratio is further applied to retain larger bounding boxes.

Overall, the near-view model consistently distinguished subtle phenotypic differences that were difficult to preserve in far-view images.

Table 3

Performance of YOLO26l trained on different datasets for near-view flower detection.

Datasets	<i>imgsz</i>	<i>P</i>	<i>R</i>	mAP	Speed (ms/image)	Instances
D _{NV}	640	0.911	0.891	0.918	28.0	10,333
	960	0.920	0.910	0.932	28.2	
	1280	0.927	0.905	0.935	29.0	

Table 4

Specific results of YOLO26l trained on D_{NV} with an *imgsz* of 1280.

Classes	<i>P</i>	<i>R</i>	<i>AP</i>	mAP
BD	0.956	0.961	0.973	0.935
HOF	0.874	0.841	0.886	
OUF	0.908	0.888	0.922	
OPF	0.948	0.961	0.972	
WF	0.954	0.919	0.958	
OCF	0.925	0.858	0.907	
FC	0.938	0.941	0.967	
BJ	0.912	0.874	0.899	
All	0.927	0.905	-	

In contrast, the far-view model provided broader canopy coverage but was more sensitive to reduced flower scale and foliage interference. A single-view configuration may therefore be sufficient when either



Fig. 7. Near-view detection examples from YOLO26l, where overlapping FC detection boxes are marked with an orange circle. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

canopy-level flower distribution or local flower classification alone meets the operational requirement. Dual-view perception is more appropriate when broad workspace observation and fine-grained flower-state assessment are required simultaneously. Under the tested trellised canopy conditions, the far-view and near-view imaging distances of approximately 1.4 m and 0.5 m, respectively, provided these complementary observation scales. These distances should be regarded as reference settings for the present cameras because their transfer to another system depends on camera FoV, flower pixel size, canopy density, and the required spatial coverage. Establishing correspondence between the two views is consequently necessary for maintaining flower cluster identity across observation scales.

3.3. Performance of flower cluster matching

The proposed cross-view flower cluster matching method established stable correspondences between far-view and near-view images, enabling effective spatial association across different viewpoints. The matching performance was evaluated on an independent set of 30 image pairs, separate from the training and validation datasets, using a deployed edge-computing platform (Jetson AGX Orin, NVIDIA, USA), with 448 FC detected in the near-view images. The overall matching reached an Acc of $94.83\% \pm 6.83\%$ for GDM strategy, with a corresponding 95% confidence interval of [92.39%, 97.27%], as summarised in Table 5. In comparison, LDM achieved an Acc of $86.94\% \pm 11.56\%$ with a wider confidence interval of [82.80%, 91.08%]. Although LDM offered higher computational efficiency, it introduced greater geometric uncertainty, which could lead to imprecise homography transformations and noticeable positional drift. The paired *t*-test showed that the improvement achieved by GDM was statistically significant ($p = 4.53 \times 10^{-5} < 0.001$), indicating that the performance gain was not caused by random variation. Beyond this statistical improvement, the GDM accuracy corresponds to an average association error rate of 5.17% under the tested conditions. In a subsequent robotic workflow, such

Table 5
Performances of flower cluster matching by different strategies.

Performances		Matching strategies	
		GDM	LDM
Acc		$94.83\% \pm 6.83\%$	$86.94\% \pm 11.56\%$
95% Confidence intervals		[92.39%, 97.27%]	[82.80%, 91.08%]
Paired <i>t</i> -test		$p = 4.53 \times 10^{-5} (***)$	
Average matching speed (s/images pair)	Object detection	0.142	
	Homography estimation	0.649	0.394
	Descriptor construction	0.038	
	Similarity calculation	0.020	
	Total	0.849	0.594

Note: ***, $p < 0.001$.

errors may lead to omitted target transfer, repeated target selection, or incorrect transfer of flower status between cross-views, rather than the same proportional pollination failure rate.

The computational analysis further demonstrated that homography estimation represented the primary processing bottleneck, requiring 0.649 s for GDM and 0.394 s for LDM, whereas object detection, descriptor construction, and similarity calculation introduced relatively limited computational overhead. Consequently, the total average processing time was 0.849 s per image pair for GDM and 0.594 s for LDM. Comparable perception times have been documented in recent agricultural robotic studies. The apple-tree pruning robot system reported by Tong et al. (2026) required an average time of 6.1 s for pruning point identification and 17.8 s for a complete branch operation on a notebook equipped with an Nvidia GTX 3050 GPU. In addition, the strawberry harvesting robot developed by He, Liu, et al. (2025) required 6.26 s for an integrated machine-vision process comprising camera activation,

data acquisition, and image processing on a laptop equipped with an Nvidia RTX 3070 GPU. Although differences in perception tasks, hardware configurations, and timing definitions prevent direct comparison, these results suggest that the 0.849 s required by GDM is computationally acceptable as part of an integrated agricultural robotic perception process.

GDM achieved more stable geometric matching than LDM because it incorporated broader canopy structure information rather than relying only on local floral features. Although LDM offered higher computational efficiency by restricting feature extraction to local flower regions, its limited spatial support introduced greater geometric uncertainty, which could lead to imprecise homography transformations and noticeable positional drift, as illustrated in Fig. 8a. In contrast, GDM leveraged the structural continuity of branches and the textural distribution of leaves to provide abundant and stable geometric cues, effectively mitigating positional drift arising from unstable homography estimation based only on local floral features, as shown in Fig. 8b. The incorporation of broader structural and textural background information also reduced sensitivity to local illumination fluctuation and partial shadowing commonly observed in field orchards. Nevertheless, homography estimation still assumes an approximately planar canopy structure, and excessive canopy thickness caused by non-standardised pruning may introduce residual geometric errors under practical orchard conditions.

Further analysis of failed matching cases revealed that errors can propagate through the detection to matching process. In far-view, low pixel density, partial occlusion, and visual similarity between unopened flowers and leaves can lead to missed or false detections, which affect the completeness and reliability of the canopy-level candidate set. In near-view, dense floral growth, overlapping branches, and unclear cluster boundaries can cause adjacent flower clusters to be merged into a single detection box, as shown in Fig. 7. As the matching process constructs positional and appearance descriptors from detected flower

cluster boxes, such detection errors directly alter the candidate number, bounding-box centre, and local colour distribution. These descriptor distortions reduce positional similarity after homography transformation and appearance similarity during cross-view matching, eventually leading to omitted or incorrect flower cluster correspondences, as shown in Fig. 9. Despite these challenges, the proposed cluster feature construction and matching scheme demonstrated strong robustness, maintaining reliable correspondences even under local misalignment and ensuring stable matching across different viewpoints. Future work should consider depth assisted registration, temporal tracking, improved separation of overlapping flower clusters, and closed loop validation of manipulator targeting as well as pollination performance under different canopy structures and flowering stages.

Compared with existing studies that primarily emphasize near-view flower detection for kiwifruit pollination, the proposed dual-scale perception framework provides both higher class diversity and competitive accuracy. Under near-view conditions, Li, Suo, et al. (2022) focused on the detection of buds and open flowers at an imaging distance of 0.5 m, achieving mAP values of 0.952 and 0.976 using YOLOv3 and YOLOv4, respectively. Li, Fu, et al. (2022) further extended to classification eleven flowering-stage categories and achieved an mAP of 0.932 using YOLOv5l. Similarly, Pan et al. (2024) reported an mAP of 0.916 for four classes flower detection using an improved YOLOv5 model, while Pinheiro et al. (2024) obtained mAP values of 0.850 and 0.940 for two classes near-view detection using YOLOv8 and DETR in near view, respectively. These studies demonstrate the effectiveness of deep learning-based flower detection under close imaging conditions, but they mainly address local flower recognition within a limited FoV. For far-view detection, Lim et al. (2020) used an improved Faster R-CNN to detect single opened flower for autonomous pollination, achieving a R of 0.798 and a P of 0.908. Although the reported results that summarized in Table 6 were obtained from different datasets, imaging distances, class definitions, model architectures, and evaluation protocols,

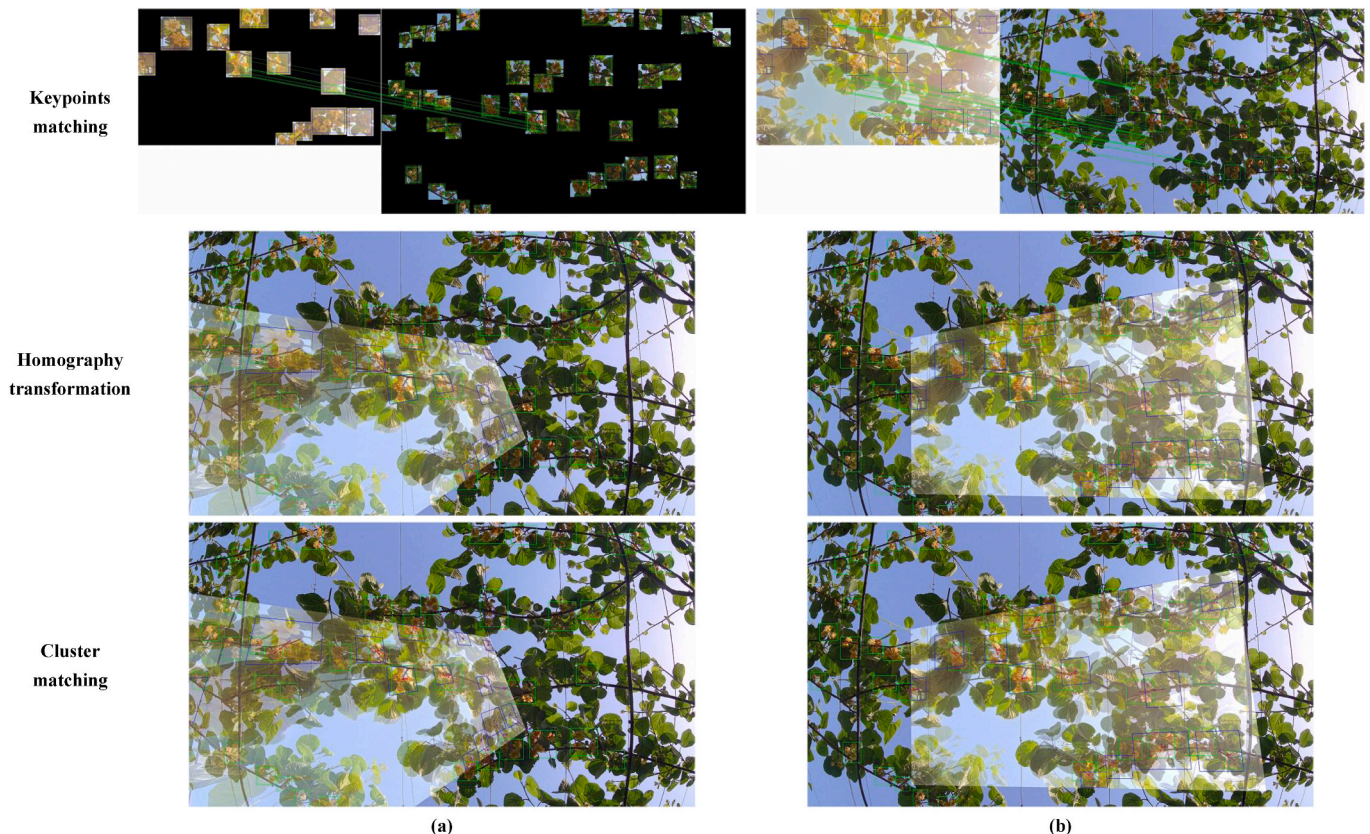


Fig. 8. Flower matching based on (a) GDM and (b) LDM, illustrating keypoints matching, homography transformation, and final cluster matching.

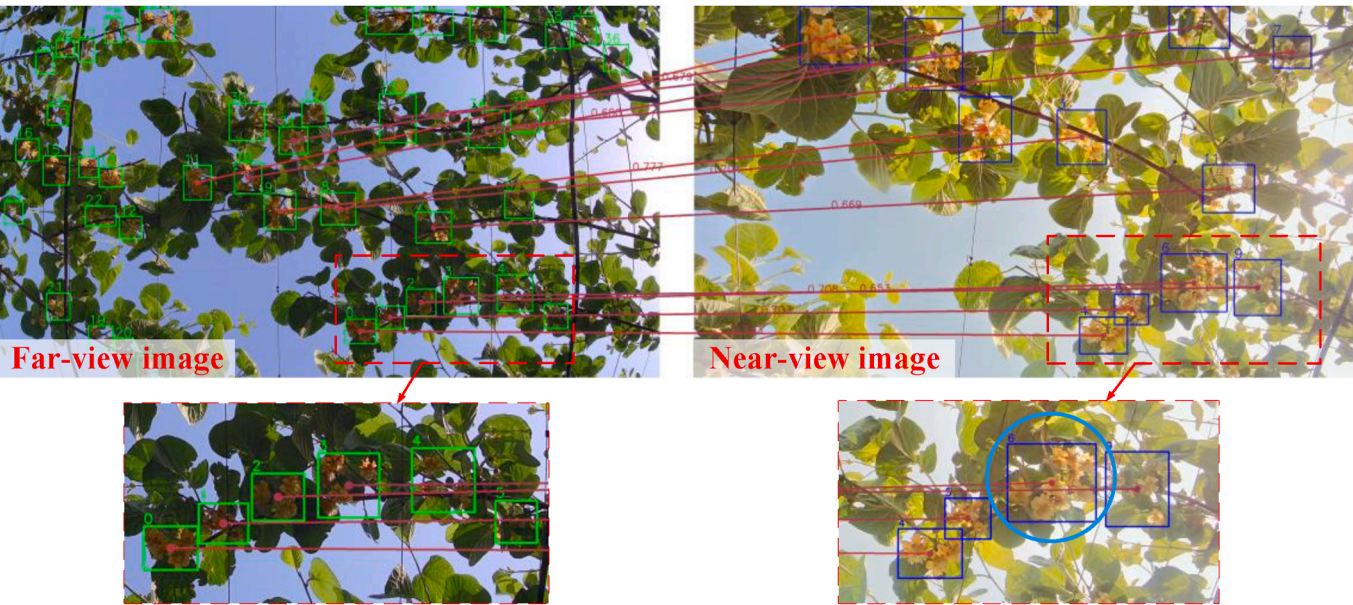


Fig. 9. Mismatches of flower cluster between far-view and near-view, which caused by mistake detections of adjacent flower clusters that marked with a blue circle. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

Table 6
Results of kiwifruit flower detection for targeted pollination.

References	Scenario	Imaging distance	Classes	Models	Evaluations
Li, Suo, et al. (2022)	Near-view	0.5 m	2	YOLOv3	mAP = 0.952
Li, Fu, et al. (2022)	Near-view	0.5 m	11	YOLOv4 YOLOv5l	mAP = 0.976 mAP = 0.932
Pan et al. (2024)	Near-view	-	4	Improved YOLOv5	mAP = 0.916
Pinheiro et al. (2024)	Near-view	-	2	YOLOv8 DETR	mAP = 0.850 mAP = 0.940
Lim et al. (2020)	Far-view	-	1	Improved Faster R-CNN	R = 0.798 P = 0.908
This study	Far-view	1.4 m	3	YOLO26l	mAP = 0.902
	Near-view	0.5 m	8	YOLO26l	mAP = 0.935

they collectively indicate that detection performance is strongly influenced by perception scale and task granularity. In general, a close-range imaging tends to reduce classification difficulty and preserve more floral details, whereas wider FoV imaging introduces smaller object scales, stronger background interference, more frequent foliage occlusion, and higher inter-class ambiguity. Following this principle, the present framework of this study used far-view detection to capture three classes flowe at 1.4 m with an mAP of 0.902, and near-view detection to recognise eight fine-grained classes at 0.5 m with an mAP of 0.935. Beyond detection accuracy, the proposed framework lies in the integration of canopy observation from the far view, detailed recognition of flower states from the near view, and homography guided flower cluster association, which linking canopy target distribution with local flower state information for robotic pollination.

4. Conclusions

This study presented a dual-scale perception framework for flower detection and spatial matching to overcome the limitations of single-view approaches in robotic kiwifruit pollination. Far view detection preserved canopy coverage and provided candidate information on

flower clusters and major flowering stages, although small target size, background interference, and occlusion limited its ability to support detailed flower state judgement. Near view detection complemented this limitation by retaining richer floral morphology and enabling more refined recognition of flower development, occlusion, pollination status, cluster structure, and branch junctions. The comparison of input configurations further indicated that increasing image detail improves recognition only until the available spatial information becomes sufficient, while flower morphology variation and dense growth remain important sources of uncertainty. For cross view association, using broader canopy structures helped reduce homography drift, and the hybrid position and appearance descriptors improved the stability of flower cluster matching. Although the framework provides a practical basis for target selection and task planning, its field deployment still depends on relatively standardised canopy structures and remains affected by illumination variation, occlusion, and residual geometric errors. Future work should integrate depth information, temporal tracking, and closed loop robotic validation under diverse orchard conditions.

Data availability

Data is available at <https://github.com/Smart-Ag/Far-Near-kiwiflower-detection-matching-images>.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT and DeepSeek for grammar checking to improve the readability of article. After using these servers, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

CRedit authorship contribution statement

Leilei He: Data curation, Investigation, Methodology, Writing – original draft. Bo Jia: Conceptualization, Data curation, Methodology, Writing – review & editing. Juncal Huang: Conceptualization, Methodology, Writing – review & editing. Rui Li: Conceptualization,

Methodology, Writing – review & editing. **Spyros Fountas**: Conceptualization, Methodology, Writing – original draft. **Longsheng Fu**: Conceptualization, Methodology, Supervision, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was partially supported by National Natural Science Foundation of China (32371999, 32501786); Chinese Universities Scientific Fund (2452026021); National Foreign Expert Project, Ministry of Human Resources and Social Security, China (H20240238, Y20240046); Key Research and Development Program of Shaanxi, China (2024 PT-ZCK-27); Key Research and Development Program of Shaanxi, China (2024NC-YBXM-195); Central Public-interest Scientific Institution Basal Research Fund, China (1610192023105); National Key Research and Development Program, China (2022YFD1600700).

References

- Abbate, A. P., Campbell, J. W., Vinson, E. L., & Williams, G. R. (2021). The pollination and fruit quality of two kiwifruit cultivars (*Actinidia chinensis* Var. *chinensis* 'AU Golden Sunshine' and 'AU gulf Coast gold') (ericales: Actinidiaceae) grown in the southeastern United States. *Journal of Economic Entomology*, 114(3), 1234–1241. <https://doi.org/10.1093/jeet/toab075>
- Akyon, F. C., Altinuc, S. O., & Temizel, A. (2022). Slicing aided hyper inference and finetuning for small object detection. In *2022 IEEE international conference on image processing (ICIP)* (pp. 966–970). IEEE. <https://doi.org/10.1109/ICIP46576.2022.9897990>.
- Alyafei, M. A. S., Dakheel, A. A., Almoosa, M., & Ahmed, Z. F. R. (2022). Innovative and effective spray method for artificial pollination of date palm using drone. *HortScience*, 57(10), 1298–1305. <https://doi.org/10.21273/HORTSCI16739-22>
- Bai, Y., Yu, J., Yang, S., & Ning, J. (2024). An improved YOLO algorithm for detecting flowers and fruits on strawberry seedlings. *Biosystems Engineering*, 237, 1–12. <https://doi.org/10.1016/j.biosystemseng.2023.11.008>
- Dang, H., He, L., Shi, Y., Jannet, L. L., Liu, X., Chen, C., Li, R., Ye, H., Chen, J., Majeed, Y., Kou, X., & Fu, L. (2025). Growth characteristics based multi-class kiwifruit bud detection with overlap-partitioning algorithm for robotic thinning. *Computers and Electronics in Agriculture*, 229, Article 109715. <https://doi.org/10.1016/j.compag.2024.109715>
- Diaz-Ramirez, V. H., Gonzalez-Ruiz, M., Juarez-Salazar, R., & Cazorla, M. (2025). Reliable disparity estimation using multiscalar vision with adjustable baseline. *Sensors*, 25(1), 21. <https://doi.org/10.3390/s25010021>
- Du, Y., Han, Y., Su, Y., & Wang, J. (2024). A lightweight model based on you only look once for pomegranate before fruit thinning in complex environment. *Engineering Applications of Artificial Intelligence*, 137, Article 109123. <https://doi.org/10.1016/j.engappai.2024.109123>
- Elbasi, E., Topcu, A. E., Cina, E., Zreikat, A. I., Shdefat, A., Zaki, C., & Abdelbaki, W. (2024). Enhanced plant leaf classification over a large number of classes using machine learning. *Applied Sciences*, 14(22), Article 10507. <https://doi.org/10.3390/app142210507>
- Everingham, M., Van Gool, L., Williams, C. K. I., Winn, J., & Zisserman, A. (2010). The pascal visual object classes (VOC) challenge. *International Journal of Computer Vision*, 88(2), 303–338. <https://doi.org/10.1007/s11263-009-0275-4>
- Gao, C., He, L., Ding, Y., Murengami, B. G., Chen, J., Zhou, C., Ye, H., Li, R., & Fu, L. (2025). A novel multinozzle targeting pollination robot for clustered kiwifruit flowers based on air-liquid dual-flow spraying. *Journal of Field Robotics*, 42(5), 2136–2150. <https://doi.org/10.1002/rob.22499>
- Gao, C., He, L., Fang, W., Wu, Z., Jiang, H., Li, R., & Fu, L. (2023). A novel pollination robot for kiwifruit flower based on preferential flowers selection and precisely target. *Computers and Electronics in Agriculture*, 207, Article 107762. <https://doi.org/10.1016/j.compag.2023.107762>
- Garibaldi, L. A., Steffan-Dewenter, I., Winfree, R., Aizen, M. A., Bommarco, R., Cunningham, S. A., Kremen, C., & Carvalheiro, L. G. (2014). Wild pollinators enhance fruit set of crops regardless of honey bee abundance. *Science*, 339(6217), 1608–1611. <https://doi.org/10.1126/science.1230200>
- He, L., Fang, W., Zhao, G., Wu, Z., Fu, L., Li, R., Majeed, Y., & Dhupia, J. (2022). Fruit yield prediction and estimation in orchards: A state-of-the-art comprehensive review for both direct and indirect methods. *Computers and Electronics in Agriculture*, 195, Article 106812. <https://doi.org/10.1016/j.compag.2022.106812>
- He, L., Wei, R., Ding, Y., Huang, J., Wei, X., Li, R., Wang, S., & Fu, L. (2025). Enhancing dense-scene millet appearance quality inspection based on YOLO11s with overlap-partitioning strategy for procurement. *Agronomy*, 15(6), 1284. <https://doi.org/10.3390/agronomy15061284>
- He, Z., Liu, Z., Zhou, Z., Karkee, M., & Zhang, Q. (2025). Improving picking efficiency under occlusion: Design, development, and field evaluation of an innovative robotic strawberry harvester. *Computers and Electronics in Agriculture*, 237, Article 110684. <https://doi.org/10.1016/j.compag.2025.110684>
- Hidayatullah, P., Tubagus, R., & Bandung, P. N. (2026). YOLO26: A comprehensive architecture overview and key improvements. *arXiv preprint*, arXiv: 2602.14582Vol. 1.
- Hiraguri, T., Shimizu, H., Kimura, T., Matsuda, T., Maruta, K., Takemura, Y., Ohya, T., & Takanashi, T. (2023). Autonomous drone-based pollination system using AI classifier to replace bees for greenhouse tomato cultivation. *IEEE Access*, 11, 99352–99364. <https://doi.org/10.1109/ACCESS.2023.3312151>
- Leutenegger, S., Chli, M., & Siegwart, R. Y. (2011). BRISK: Binary robust invariant scalable keypoints. In *2011 IEEE international conference on computer vision (ICCV)* (pp. 2548–2555). IEEE. <https://doi.org/10.1109/ICCV.2011.6126542>
- Li, G., Fu, L., Gao, C., Fang, W., Zhao, G., Shi, F., Dhupia, J., Zhao, K., Li, R., & Cui, Y. (2022). Multi-class detection of kiwifruit flower and its distribution identification in orchard based on YOLOv5l and euclidean distance. *Computers and Electronics in Agriculture*, 201, Article 107342. <https://doi.org/10.1016/j.compag.2022.107342>
- Li, G., Suo, R., Zhao, G., Gao, C., Fu, L., Shi, F., Dhupia, J., Li, R., & Cui, Y. (2022). Real-time detection of kiwifruit flower and bud simultaneously in orchard using YOLOv4 for robotic pollination. *Computers and Electronics in Agriculture*, 193, Article 106641. <https://doi.org/10.1016/j.compag.2021.106641>
- Lim, J. Y., Ahn, H. S., Nejadi, M., Bell, J., Williams, H., & MacDonald, B. A. (2020). Deep neural network based real-time kiwi fruit flower detection in an orchard environment. *arXiv preprint*, arXiv:2006.04343.
- Lin, T. Y., Maire, M., Belongie, S., Hays, J., Perona, P., Ramanan, D., Dollár, P., & Zitnick, C. L. (2014). Microsoft COCO: Common objects in context. In D. Fleet, T. Pajdla, B. Schiele, & T. Tuytelaars (Eds.), *Computer vision – ECCV 2014. Lecture notes in computer science* (Vol. 8693, pp. 740–755). Springer. https://doi.org/10.1007/978-3-319-10602-1_48
- Liu, J., & Li, S. (2025). Method for detecting densely distributed kiwifruit flowering period based on improved YOLOv8n. *Transactions of the Chinese Society of Agricultural Engineering*, 41(12), 172–181. <https://doi.org/10.11975/j.issn.1002-6819.202502138>
- Magalhães, S. A., Moreira, A. P., Santos, F. N., & Dias, J. (2022). Active perception fruit harvesting robots — A systematic review. *Journal of Intelligent and Robotic Systems*, 105(1), 14. <https://doi.org/10.1007/s10846-022-01595-3>
- Mazinani, M., Zarafshan, P., Dehghani, M., Vahdati, K., & Etezadi, H. (2023). Design and analysis of an aerial pollination system for walnut trees. *Biosystems Engineering*, 225, 83–98. <https://doi.org/10.1016/j.biosystemseng.2022.12.001>
- Millard, J., Outhwaite, C. L., Ceaşu, S., Carvalheiro, L. G., da Silva e Silva, F. D., Dicks, L. V., Ollerton, J., & Newbold, T. (2023). Key tropical crops at risk from pollinator loss due to climate change and land use. *Science Advances*, 9(41). <https://doi.org/10.1126/sciadv.adh0756>
- Mirhaji, H., Soleymani, M., Asakereh, A., & Abdanan Mehdizadeh, S. (2021). Fruit detection and load estimation of an Orange orchard using the YOLO models through simple approaches in different imaging and illumination conditions. *Computers and Electronics in Agriculture*, 191, Article 106533. <https://doi.org/10.1016/j.compag.2021.106533>
- Pan, F., Hu, M., Duan, X., Zhang, B., Xiang, P., Jia, L., Zhao, X., & He, D. (2024). Enhancing kiwifruit flower pollination detection through frequency domain feature fusion: A novel approach to agricultural monitoring. *Frontiers in Plant Science*, 15, Article 1415884. <https://doi.org/10.3389/fpls.2024.1415884>
- Pandey, P., Acosta, J. J., Payn, K. G., & Young, S. (2024). Towards autonomous, aerial pollination: Design of a robotic pollinator payload for controlled crosses in loblolly pine. *Applied Engineering in Agriculture*, 40(6), 635–649. <https://doi.org/10.13031/aea.15916>
- Pinheiro, I., Moreira, G., Magalhães, S., Valente, A., Cunha, M., & Dos Santos, F. N. (2024). Deep learning based approach for actinidia flower detection and gender assessment. *Scientific Reports*, 14, Article 24452. <https://doi.org/10.1038/s41598-024-73035-1>
- Rodger, J. G., Bennett, J. M., Razanajatovo, M., Knight, T. M., van Kleunen, M., Ashman, T. L., Steets, J. A., Hui, C., Arceo-Gómez, G., Burd, M., Burkle, L. A., Burns, J. H., Durka, W., Freitas, L., Kemp, J. E., Li, J., Pauw, A., Vamosi, J. C., Wolowski, M., Xia, J., & Ellis, A. G. (2021). Widespread vulnerability of flowering plant seed production to pollinator declines. *Science Advances*, 7(42). <https://doi.org/10.1126/sciadv.abd3524>
- Rybak, L. A., Cherkasov, V. V., & Voloshin, A. A. (2025). Two stage technical vision algorithm for automatic fruit picking based on neural networks and hough transform. In *2025 international conference on industrial engineering, applications and manufacturing (ICIEAM)* (pp. 737–742). IEEE. <https://doi.org/10.1109/ICIEAM65163.2025.11028377>
- Sapkota, R., Cheppally, R. H., Sharda, A., & Karkee, M. (2026). YOLO26: Key architectural enhancements and performance benchmarking for real-time object detection. *arXiv preprint*, arXiv: 2509.25164Vol. 5.
- Tong, S., Wang, J., Zhang, J., Liu, L., Wang, Y., & Kang, F. (2026). An Apple tree pruning robot system based on branch segmentation and decision-making control. *Artificial Intelligence in Agriculture*, 16, 1025–1040. <https://doi.org/10.1016/j.aiaa.2026.04.005>
- Uddin, Z., Mahamood, N., Ray, A., Pramanik, I., & Alnajjar, F. (2026). E2ETCA: End-to-end training of CNN and attention ensembles for rice disease diagnosis. *Journal of Integrative Agriculture*, 25(4), 756–768. <https://doi.org/10.1016/j.jia.2024.03.075>
- Villacrés, J., & Vougioukas, S. (2024). Assessing a multi-camera system to enhance fruit visibility for robotic harvesting in a V-trellised Apple orchard. *Computers and*

- Electronics in Agriculture*, 224, Article 109164. <https://doi.org/10.1016/j.compag.2024.109164>
- Wang, C., Yang, M., Han, Q., Li, P., He, Z., Zhao, Z., & Wu, F. (2026). Dynamic litchi fruit trajectory prediction using extended kalman filter in simulated wind environment. *Computers and Electronics in Agriculture*, 247, Article 111666. <https://doi.org/10.1016/j.compag.2026.111666>
- Wang, H., Lin, Y., Xu, X., Chen, Z., Wu, Z., & Tang, Y. (2022). A study on long-close distance coordination control strategy for litchi picking. *Agronomy*, 12(6), 1250. <https://doi.org/10.3390/agronomy12071520>
- Wang, H., Zhang, G., Cao, H., Hu, K., Wang, Q., Deng, Y., Gao, J., & Tang, Y. (2025). Geometry-aware 3D point cloud learning for precise cutting-point detection in unstructured field environments. *Journal of Field Robotics*, 42(7), 3063–3076. <https://doi.org/10.1002/rob.22567>
- Wang, X., Burusa, A. K., Rapado-Rincón, D., Versmissen, T. A. W., van Henten, E. J., & Kootstra, G. (2025). Multiple object tracking with multi-view active vision to effectively find plant nodes in a cluttered tomato greenhouse. *Computers and Electronics in Agriculture*, 234, Article 110266. <https://doi.org/10.1016/j.compag.2025.110266>
- Williams, H., Nejati, M., Hussein, S., Penhall, N., Lim, J. Y., Jones, M. H., Bell, J., Ahn, H. S., Bradley, S., Schaare, P., Martinsen, P., Alomar, M., Patel, P., Seabright, M., Duke, M., Scarfe, A., & MacDonald, B. (2020). Autonomous pollination of individual kiwifruit flowers: Toward a robotic kiwifruit pollinator. *Journal of Field Robotics*, 37(2), 246–262. <https://doi.org/10.1002/rob.21861>
- Xiao, C., Wang, Y., Zhao, H., Eneji, A. E., Xu, D., Du, M., Du, X., Li, Q., Tian, X., & Li, Z. (2025). A method for rapid field monitoring of boll opening rate for cotton based on digital imagery. *Journal of Integrative Agriculture*. <https://doi.org/10.1016/j.jia.2025.07.018> (Article in press).
- Xie, Y., Zhang, T., Yang, M., Lyu, H., Zou, Y., Sun, Y., Xiao, J., Lian, W., Tao, J., Han, H., & Xu, C. (2025). Engineering crop flower morphology facilitates robotization of cross-pollination and speed breeding. *Cell*, 188(21), 5809–5830.e27. <https://doi.org/10.1016/j.cell.2025.07.028>
- Xu, T., Li, X., Qi, Y., Zeng, J., & Lv, X. (2023). Experiment on liquid pollination of pear tree with horizontal scaffolding based on multi-rotor UAV. *Transactions of the Chinese Society for Agricultural Machinery*, 54(S2), 136–141. <https://doi.org/10.6041/j.issn.1000-1298.2023.S2.015>
- Xu, Y., Li, B., Zhai, Y., Ke, F., Zhou, J., Coscia, P., Genovese, A., Piuri, V., & Scotti, F. (2025). PBSNet: Prismatic battery surface defect detection via sliding slice amplification and shunted dynamic snake convolution. *IEEE Transactions on Automation Science and Engineering*. <https://doi.org/10.1109/TASE.2025.3620333>
- Yang, J., Li, X., Wang, X., Fu, L., & Li, S. (2024). Vision-based localization method for picking points in tea-harvesting robots. *Sensors*, 24(21), 6777. <https://doi.org/10.3390/s24216777>
- Yang, L., Noguchi, T., & Hoshino, Y. (2024). Development of a grape cut point detection system using multi-cameras for a grape-harvesting robot. *Sensors*, 24(24), 8035. <https://doi.org/10.3390/s24248035>
- Zhang, Q., Zhang, Z., Manzoor, S. H., Li, T., Igathinathane, C., Li, W., Zhang, M., Mhamed, M., Javidan, S. M., & Abdelhamid, M. (2025). A comprehensive review of autonomous flower pollination techniques: Progress, challenges, and future directions. *Computers and Electronics in Agriculture*, 237, Article 110577. <https://doi.org/10.1016/j.compag.2025.110577>
- Zhang, W., Song, F., & Sun, J. (2026). A review of strategic enhancement of pollination with smart agriculture to counteract the decline of natural pollinators. *Biosystems Engineering*, 261, Article 104344. <https://doi.org/10.1016/j.biosystemseng.2025.104344>
- Zhou, G., Ma, S., & Liang, F. (2022). Recognition of the Apple in panoramic images based on improved YOLOv4 model. *Transactions of the Chinese Society of Agricultural Engineering*, 38(21), 159–168. <https://doi.org/10.11975/j.issn.1002-6819.2022.21.019>
- Zhu, Y., Sui, S., Du, W., Li, X., & Liu, P. (2025). Picking point localization method of table grape picking robot based on you only look once version 8 nano. *Engineering Applications of Artificial Intelligence*, 146, Article 110266. <https://doi.org/10.1016/j.engappai.2025.110266>